



Indian Institute of Technology Kanpur
Department of Mathematics and Statistics
Introduction to Bayesian Analysis (MTH422)
Quiz 2, Date: February 14, 2025, Friday

Time: 8:00–8:40 AM (8:00–8:55 AM for DAP students)

Spring 2025

Max point: 10

The answers need to be explicit. Writing some philosophical answer and claiming that it is correct will lead to no point. Calculation mistakes will be penalized.

1. Suppose $X_1 \sim \text{Normal}(0, 1)$ and $X_t|X_{t-1} \sim \text{Normal}(\rho X_{t-1}, 1 - \rho^2)$, $t = 2, \dots, T$. Find the joint distribution of $\mathbf{X} = (X_1, \dots, X_T)'$. Write down the conditional distribution of X_t given $X_1, \dots, X_{t-1}, X_{t+1}, \dots, X_T$ for some $t \in \{2, 3, \dots, T-1\}$. A closed-form analytical expression is required for the second part. Writing final answers just as proportional to some quantities is not allowed. (1 + 2 = 3 points)
2. Suppose $Z_t|Y_t \stackrel{\text{Indep}}{\sim} \text{Normal}(Y_t, 2^2)$, $t = 1, \dots, T$, and $Y_1, \dots, Y_T|\lambda \stackrel{\text{iid}}{\sim} \text{Poisson}(\lambda)$, and $\lambda \sim \text{Gamma}(a, b)$. Using Bayes theorem, obtain the conditional distribution of λ given $Y_1, \dots, Y_T$. (1.5 points)
3. Say $Y|\lambda \sim \text{Poisson}(\lambda)$. (a) Derive and plot the Jeffreys' prior for λ . (b) Is this prior proper? (c) Derive the posterior and give conditions on Y to ensure it is proper. (1+0.5+1=2.5 points)
4. Suppose the likelihood $f(Y|\theta) \geq c(Y)$ for all θ . Show that an improper prior on θ will yield an improper posterior. (1.5 points)
5. Show that the mixture of experts prior is a conjugate when the prior components are conjugate. (1.5 points)